Lesson Plan: Decoding Enzyme-Substrate Mechanics

Grade Level: High School Biology (Honors Biology)
Duration: 50 Minutes

Lesson Overview & Standards Alignment
Next Generation Science Standards (NGSS):
Performance Expectation:HS-LS1-2: Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.
Science & Engineering Practices (SEP):
Developing and Using Models: Use a model based on evidence to illustrate the relationships between systems or components.
Analyzing and Interpreting Data: Analyze data using tools/models to determine similarities and differences in findings.
Disciplinary Core Ideas (DCI):
LS1.A: Structure and Function: Systems of specialized cells within organisms help them perform the essential functions of life.
Crosscutting Concepts (CCC):
Structure and Function: The functions and properties of natural and designed objects are determined by their sub-structures and molecular organization.
Stability and Change: Feedback mechanisms can encourage or discourage changes in systems.

Learning Objectives:
Students will be able to explain the Induced Fit Model versus the classic Lock and Key Hypothesis.
Students will be able to predict and observe how temperature and pH changes cause denaturation.

Materials Needed
Station 1 (Printables): Paper cut-outs of enzymes (with unique active site shapes) and substrates.
Station 2 (Physical Models): 6–8 different padlocks with matching physical keys (or **3D-printed puzzle pieces *I LOVE THIS IDEA!***).
Digital Access: Laptops/Tablets running the Enzyme 3D Visualization App.

Student Handout: “Enzyme Mechanics Lab Sheet”

Activity Schedule
Phase 1: Paper Cut-Out Match (15 Mins)
Setup: Place mixed paper cut-outs of 4 different enzymes (e.g., Hexokinase, Adenylate Kinase, DNA Polymerase, Maltose Binding Protein) and 6–8 substrate candidates on each student lab table.
Task: Working in pairs, students must attempt to align substrates with enzyme active sites.
Key Concept Highlight: Students will notice that while some paper cut-outs fit instantly (Lock & Key), others look like they almost fit, but need a structural shift (Induced Fit).

Phase 2: "Find Your Specific Substrate" Human Lock-and-Key Game (15 Mins)
Setup: Hand half the class a labeled physical lock (Enzyme) and the other half a labeled key (Substrate).
The Rules:
Students move quietly around the classroom trying keys in locks to find their matching pair.
The Twist: 2–3 keys should be slightly taped or physically altered (simulating a substrate that requires conformational change or a competitive inhibitor).
Once students have found their matching key set, students must draw the locks and keys of their matching set and at least one other matching set.
Reflection Question: "When you found your matching key, did it open effortlessly, or did you have to apply torque/force to turn it?" (This introduces the idea that binding causes structural stress on bonds).

Phase 3: Interactive 3D Visualization Lab (25 Mins)
Students transition to the digital app to observe real PDB structural data at the molecular level.
Part A: Open vs. Closed Conformation (Induced Fit)
Select Hexokinase and switch between Open State and Closed State.
Prompt: "Describe what happens to the two lobes of Hexokinase as it captures Glucose."
Switch the display to VDW Surface Envelope.
Prompt: "Why is the Van der Waals surface helpful for seeing how the substrate is engulfed compared to the cartoon ribbon view?"
Part B: Environmental Stress & Denaturation
Pick an enzyme and slowly raise the Temperature slider past its optimal limit (50C).
Prompt: "What happens to the 3D backbone fold when the enzyme denatures? Why can't the substrate fit anymore?"
Adjust the pH slider to extreme acid (pH 2) or extreme base (pH 12).
Prompt: "How does changing pH alter the functional shape of the protein differently than heat?"